

FACTSHEET FS11 · Residual momentum

What changed on 2 October 2026. Two corrections, both to the universe of stocks, and every number below is re-run on them. (1) **US:** the universe is now the actual S&P 500 members at each month-end. Until now it was the 500 largest of every company that was *ever* in the index 2012–2026, which includes later winners before they joined. (2) **UK and EU:** the price panels now include the index members that stopped trading (UK coverage of members 71% → 92%, EU 81% → 89%); before, most of them had no prices. SCANDI stays as registered (OMXC25 + OMXS30 + OMXH25). Registered verdicts are unchanged; where a corrected number crosses a gate, the text says so. Comparison and method: FACTSHEETS_RERUN_2026-10-02.md.

US, 1999–2012 (pre-registered second period, run 2 October 2026). The same rules on Sharadar's point-in-time S&P 500 with delisted stocks, 1999–2012, a period this factsheet was not built on. L/S net: t -0.12 (2013–2026: +1.22), reversed. Long-only alpha vs the equal-weight universe: +0.1% a year, t +0.05 (2013–2026: +2.30), same sign, weaker. Positive in 2000–2005, negative in 2006–2012 (the 2009 momentum crash). Registered verdicts are unchanged. Pre-registration, method and every result: PREREG_US_1999_2012.md.

Momentum without the market's ups and downs

Henrik Gjerning · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

STRATEGY	ORIGIN	FAMILY
Residual (idiosyncratic) momentum	Blitz, Huij & Martens (2011)	Momentum & trend
UNIVERSES	BACKTEST	BOOKS
US · EU · UK · DK · SCANDI · World (point-in-time)	Feb 2013 – Aug 2026, monthly, net of costs	Long/short and long-only (stocks with the strongest stock-specific trend)
REBALANCE	STATUS	
Monthly	Registered gate $ t > 2.87$ over 12 cells; every cell in the programme trial ledger	

The claim. Ranking stocks on the part of their past return that the market does not explain gives a momentum strategy with about half the volatility and far smaller crashes than plain momentum (Blitz, Huij & Martens 2011, US 1930–2009). **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned +2.2% to +6.9% a year and was positive in all six; no universe passes the 2.87 gate. The long-only book (stocks with the strongest stock-specific trend) had a higher Sharpe than the equal-weight universe in all six, with alphas of +2.4% to +3.9%, none past the gate. **What goes wrong (§9):** the main drags are a market bet (average beta -0.25, worth -2.4% a year in 2013–19), costs of about 2.0% a year. None of the pre-declared fixes passes the gate in both windows. **Classification (appendix C):** Momentum & trend; its returns sit closest to the momentum cluster of the signal library. **360° view (appendix D):** nearest neighbours Momentum 12-1, Above 52-week low; after its closest neighbours and the market it keeps an alpha of +0.8% to +6.8% (largest $|t|$ 3.5), so it carries some information of its own.

Layout revised 2 October 2026 (verdict and scorecard, Sharpe anatomy, fit with the other strategies; detail moved to the appendix). No number changed.

1. Verdict and scorecard

Verdict. Of the 12 registered cells (two books × six universes), **0 pass** the $|t| > 2.87$ gate. Programme-wide (772 trials, $|t| > 3.99$): nothing survives. US 1999–2012, a period the factsheet was not built on: L/S t -0.12, long-only alpha t +0.05.

Universe	L/S t	L/S Sharpe	Long-only alpha t	Long-only Sharpe vs equal-weight	Long-only max DD
US	+1.22	0.28	+2.30	0.90 vs 0.76	-26%
EU	+0.80	0.23	+1.88	0.86 vs 0.66	-22%
UK	+0.63	0.15	+1.22	0.68 vs 0.59	-29%
DK	+1.41	0.42	+1.60	0.80 vs 0.69	-24%
SCANDI	+0.92	0.26	+2.03	0.95 vs 0.81	-21%
World	+0.83	0.20	+1.82	0.73 vs 0.61	-25%
US 1999–2012	-0.12	-0.03	+0.05	0.46 vs 0.52	–

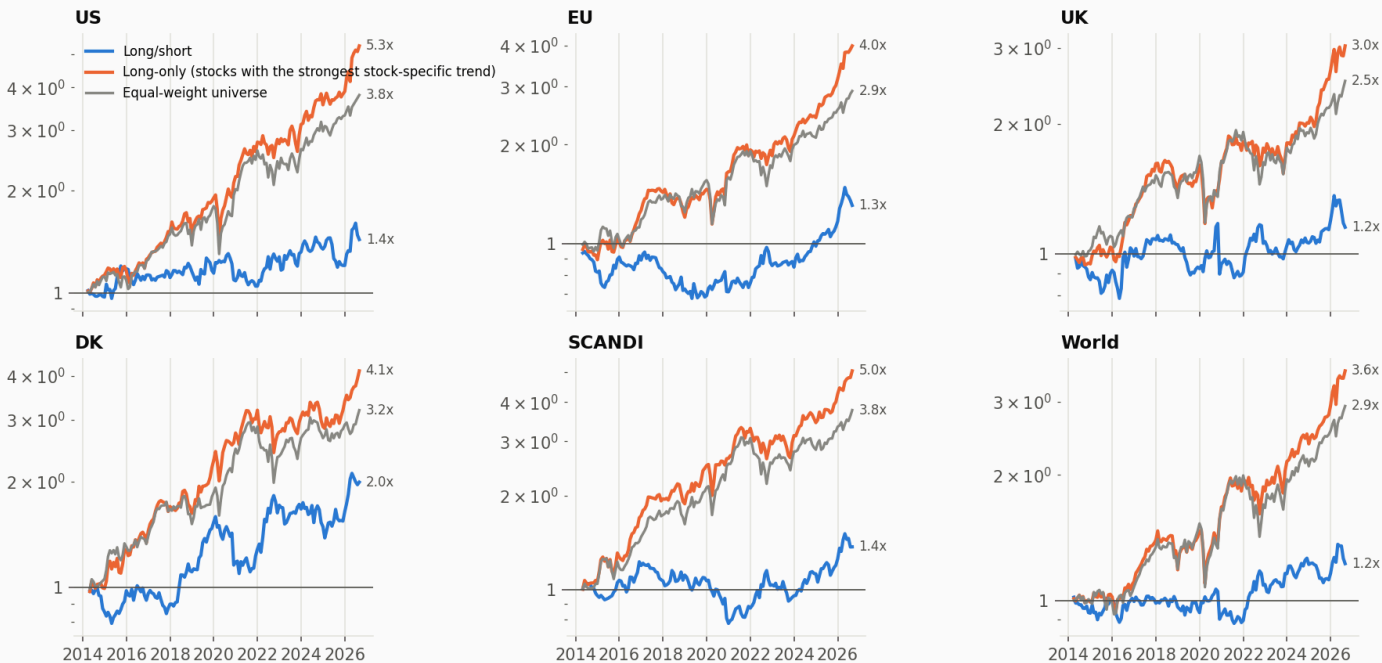
✓ passes the 2.87 gate (Bonferroni over 12 cells); ✗ significantly negative. 2013–2026 unless stated.

2. Headline performance

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long-only CAGR	Long-only Sharpe	Long-only max DD	EW CAGR	EW Sharpe	Size-weighted CAGR
US	499 / 10	+3.8%	1.22	+6.5% (2.2)	-0.22	+14.3%	0.90	-26%	+11.3%	0.76	+14.0%
EU	262 / 10	+2.9%	0.80	+5.7% (1.5)	-0.28	+11.8%	0.86	-22%	+9.0%	0.66	+9.4%
UK	319 / 10	+2.2%	0.63	+4.9% (1.4)	-0.31	+9.3%	0.68	-29%	+7.7%	0.59	+7.6%
DK	20 / 3	+6.9%	1.41	+9.8% (2.2)	-0.28	+12.1%	0.80	-24%	+9.8%	0.69	+7.8%
SCANDI	70 / 5	+3.4%	0.92	+5.8% (1.5)	-0.20	+13.9%	0.95	-21%	+11.3%	0.81	+10.5%
World	1078 / 10	+2.2%	0.83	+4.7% (2.0)	-0.24	+10.7%	0.73	-25%	+9.0%	0.61	+9.4%

Monthly, local currency (World in USD). L/S = equal-weighted stocks with the strongest stock-specific trend minus the weakest, net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West t (6 lags). CAPM vs the equal-weight universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

Residual momentum on six stock universes, 2013-2026



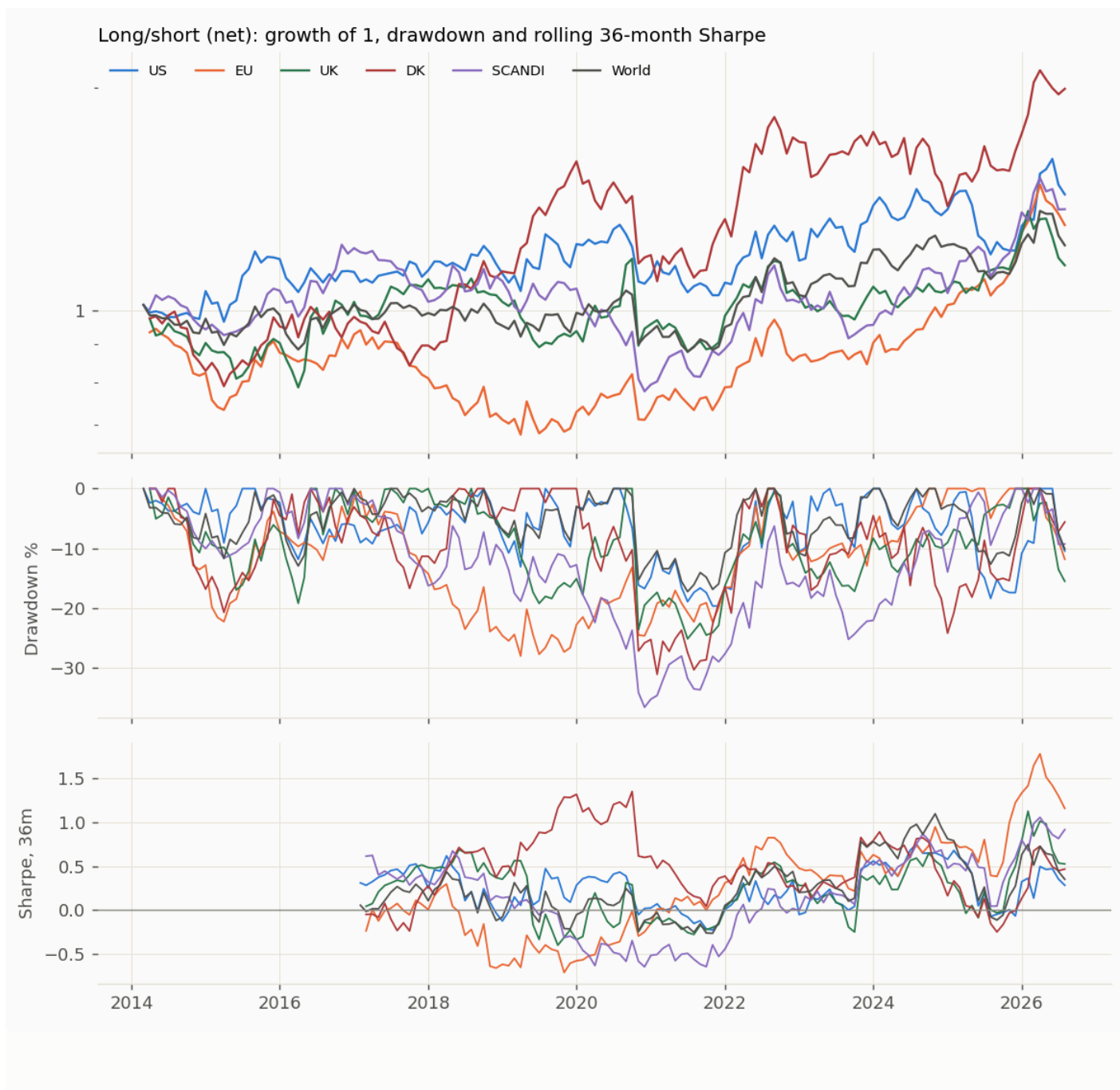
Growth of 1, log scale, local currency, net of costs. Henrik Gjernerig · Project 10 factsheets

Read with care. A long/short book with a negative beta shows a large CAPM alpha in a rising market. The factor-model alpha in section 7, after momentum, value and the other themes, is the better guide: here the L/S CAPM alpha ranges +4.7% to +9.8% and the factor alpha -4.0% to +2.0%.

3. Where the Sharpe comes from

Sharpe = annual return ÷ annual volatility. The return splits into the part the market explains (beta × market return) and the rest (alpha, net of costs); the volatility into the market's share and the strategy's own. The last column is the Sharpe the book would have with its market exposure hedged out (before hedging costs).

Universe	Book	Return / yr	= market part	+ alpha	Costs / yr (inside the return)	Volatility	of which market	Sharpe	Sharpe, market hedged
US	L/S	+3.8%	-2.6%	+6.5%	-3.1%	13.8%	3.5%	0.28	0.48
US	Long-only	+14.7%	+11.0%	+3.7%	–	16.3%	14.5%	0.90	0.51
EU	L/S	+2.9%	-2.8%	+5.7%	-3.0%	12.6%	4.2%	0.23	0.48
EU	Long-only	+12.2%	+8.3%	+3.9%	–	14.2%	12.6%	0.86	0.59



4. How it fits next to the market and the other strategies

A long/short book improves a market portfolio when its own Sharpe beats the hurdle: its correlation with the market $\times$ the market's Sharpe. A negative correlation makes the hurdle negative, so even a book with a small positive Sharpe diversifies; the size of the gain still depends on that Sharpe.

Universe	Corr. with market	Avg corr. with the other strategies	Most similar strategy	Market Sharpe	Hurdle Sharpe	This L/S Sharpe	Verdict	L/S in the worst 10% of market months	Market in those months
US	-0.25	+0.11	Momentum (+0.63)	0.76	-0.19	0.28	adds	+1.2%	-7.7%
EU	-0.33	+0.16	Momentum (+0.62)	0.66	-0.22	0.23	adds	+3.5%	-6.8%

Universe	Corr. with market	Avg corr. with the other strategies	Most similar strategy	Market Sharpe	Hurdle Sharpe	This L/S Sharpe	Verdict	L/S in the worst 10% of market months	Market in those months
UK	-0.31	+0.15	Momentum (+0.67)	0.60	-0.19	0.15	adds	+1.3%	-6.9%
DK	-0.26	+0.13	Momentum (+0.60)	0.69	-0.18	0.42	adds	+2.7%	-7.5%
SCANDI	-0.22	+0.10	Momentum (+0.57)	0.81	-0.18	0.26	adds	+3.0%	-7.2%
World	-0.36	+0.12	Momentum (+0.72)	0.61	-0.22	0.20	adds	+1.4%	-7.9%

Long/short (net), monthly, 2013–2026; market = equal-weight universe; other strategies = the long/short books of FS01–FS12. The Strategy Cockpit lets you build books of several strategies.

5. Strategy description

Ranking stocks on the part of their past return that the market does not explain gives a momentum strategy with about half the volatility and far smaller crashes than plain momentum (Blitz, Huij & Martens 2011, US 1930–2009).

Why it might work. Same under-reaction story as momentum, but removing the market (and factor) component avoids the time-varying beta that makes plain momentum crash (Grundy & Martin 2001; Daniel & Moskowitz 2016).

Rule	Original (Blitz, Huij & Martens (2011))	This factsheet
Signal	Sum of Fama-French 3-factor residuals $t-12..t-2$, over a 36-month window, divided by their volatility	Market-model residuals over 24 months, same scaling
Portfolio	Decile L/S, one month	Extreme quantile L/S, one month
Weighting	Equal weighted	Equal weighted

6. Method: data, signal and portfolio

Data load

Universe	Prices	Membership	Names eligible (median)	Currency
US	Sharadar SEP closeadj (total return), 805 tickers incl. delisted	Point-in-time: the S&P 500 members at each month-end (Sharadar add/remove history; corrected 2 Oct 2026)	~503	USD
EU	Project1 yfinance cache, Adj Close (TR)	Point-in-time: union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced	~238	local (EUR, SEK, DKK, PLN)
UK	Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×)	Point-in-time: FTSE 100 + FTSE 250, quarterly snapshots; 75% priced	~242	GBP
DK	Project1 cache, Adj Close	Point-in-time: OMX Copenhagen 25, quarterly snapshots; 91% priced	~19	DKK
SCANDI	Project1 cache, Adj Close	Point-in-time: OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced	~62	local (DKK, SEK, EUR)
World	US Sharadar + UK + EU above	Point-in-time union of the three; local-currency returns summed, not converted	~963	mixed

Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as `run_wiki_pit.py` / `run_eu_pit.py`), with a short list of verified ticker renames (e.g. `NZYM-B` → `NSIS-B`, `WDH` → `DEMANT`, `DAI` → `MBG`). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.

- **Data gap:** Market-only residuals over 24 months (the paper uses Fama-French 3 factors over 36); stock-level factor loadings outside the US would need data we do not have. The first signals appear in early 2014.

Signal creation

$\text{RESMOM}_{i,t} = \Sigma \varepsilon_{i,t-11..t-1} / \sigma(\varepsilon)$, from monthly regressions of r_i on the equal-weight universe over the last 24 months. Signals use only information up to month-end t and are traded in month $t+1$. Eligibility: in the point-in-time universe on any of the last five trading days of month t .

Model build

- Sort eligible stocks into equal-count groups (deciles ≥ 100 names, quintiles 50–99, terciles < 50 ; fixed per universe).
- **Long/short** = stocks with the strongest stock-specific trend minus the weakest; **long-only** = stocks with the strongest stock-specific trend; benchmarks: equal-weight universe and size-weighted universe.
- Equal weights within each leg, rebalanced monthly.
- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

7. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West t, 6 lags.

Long/short

Universe	Factor model	Alpha / yr	t(alpha)	R ²	Loadings with t ≥ 2 (t)
US	JKP USA 7 themes	+0.7%	0.29	0.54	value -0.29 (-2.3), momentum +0.91 (+8.7), low_risk +0.51 (+3.0), short_term_reversal +0.49 (+2.2)
EU	French Europe 5F + WML	-4.0%	-1.35	0.42	HML +0.40 (+2.4), WML +0.77 (+8.8)
UK	JKP GBR 7 themes	+0.3%	0.11	0.51	momentum +0.95 (+5.4), short_term_reversal +0.41 (+2.3)
DK	JKP DNK 7 themes	+1.0%	0.25	0.34	momentum +0.80 (+5.4), short_term_reversal +0.53 (+3.7)
SCANDI	French Europe 5F + WML	-2.2%	-0.63	0.22	HML +0.33 (+2.2), WML +0.55 (+5.5)
World	JKP World 7 themes	+2.0%	0.63	0.40	value -0.32 (-2.5), momentum +0.65 (+6.8), low_risk +0.30 (+2.3), quality -0.70 (-2.9)

Long-only

Universe	Factor model	Alpha / yr	t(alpha)	R ²	Loadings with t ≥ 2 (t)
US	JKP USA 7 themes	+3.0%	2.22	0.88	mkt +1.03 (+25.1), momentum +0.39 (+5.8), low_risk +0.32 (+2.6), quality -0.36 (-2.2)
EU	French Europe 5F + WML	+5.2%	2.06	0.62	Mkt-RF +0.71 (+13.1), HML +0.33 (+2.5)
UK	JKP GBR 7 themes	+4.0%	1.94	0.69	mkt +0.55 (+7.8), momentum +0.44 (+5.6), low_risk -0.55 (-2.6), short_term_reversal +0.37 (+2.2)
DK	JKP DNK 7 themes	+5.6%	2.22	0.59	mkt +0.67 (+9.1), size -0.29 (-2.0), short_term_reversal +0.28 (+2.4)
SCANDI	French Europe 5F + WML	+7.6%	2.14	0.49	Mkt-RF +0.67 (+8.5)
World	JKP World 7 themes	+5.7%	2.61	0.80	mkt +0.71 (+13.5), momentum +0.22 (+2.5)

8. Statistical detail

- **Gate:** |t| > 2.87 (2 books × 6 universes, Bonferroni 0.05/12, two-sided).
- **L/S mean:** positive in all six; passes in 0 of 6.
- **L/S CAPM alpha:** +4.7% to +9.8%, |t| past the gate in 0 of 6.
- **Long-only:** Sharpe above the equal-weight universe in all six; alpha +2.4% to +3.9%, passing in 0 of 6.
- **Probabilistic Sharpe** of the L/S, P(true Sharpe > 0): US 0.84, EU 0.79, UK 0.70, DK 0.93, SCANDI 0.82, World 0.76.

- **Publication decay:** gross L/S -2.6% to +10.5% a year in 2013–19 against +7.5% to +13.3% in 2020–26.

9. What goes wrong, and how to fix it

Method. Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FS03–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FS03–FS12, gate $p < 0.0020$.

9.1 Diagnosis (design window 2013–2019)

Universe	L/S gross / yr	Beta	Beta drag / yr	CAPM alpha (gross)	Costs / yr	Long leg vs EW / yr	EW vs short leg / yr	Turnover / month	Worst months
US	+6.1%	-0.41	-4.4%	+10.5%	-2.2%	+2.0%	+4.1%	91%	2015-04 -6%, 2016-02 -6%, 2019-11 -5%
EU	-2.6%	-0.34	-2.9%	+0.2%	-2.1%	-0.1%	-2.6%	89%	2018-11 -9%, 2015-02 -8%, 2014-04 -7%
UK	+2.5%	-0.33	-3.2%	+5.7%	-2.2%	+0.3%	+2.1%	94%	2016-08 -8%, 2015-06 -6%, 2019-04 -5%
DK	+10.5%	-0.06	-0.7%	+11.2%	-1.4%	+4.6%	+5.9%	59%	2014-11 -8%, 2016-04 -7%, 2016-10 -5%
SCANDI	+4.2%	-0.05	-0.8%	+4.9%	-1.8%	+3.8%	+0.3%	73%	2018-11 -6%, 2018-07 -6%, 2016-04 -5%
World	+2.4%	-0.33	-2.6%	+5.0%	-2.3%	+0.4%	+2.0%	94%	2016-02 -6%, 2019-04 -5%, 2015-02 -4%

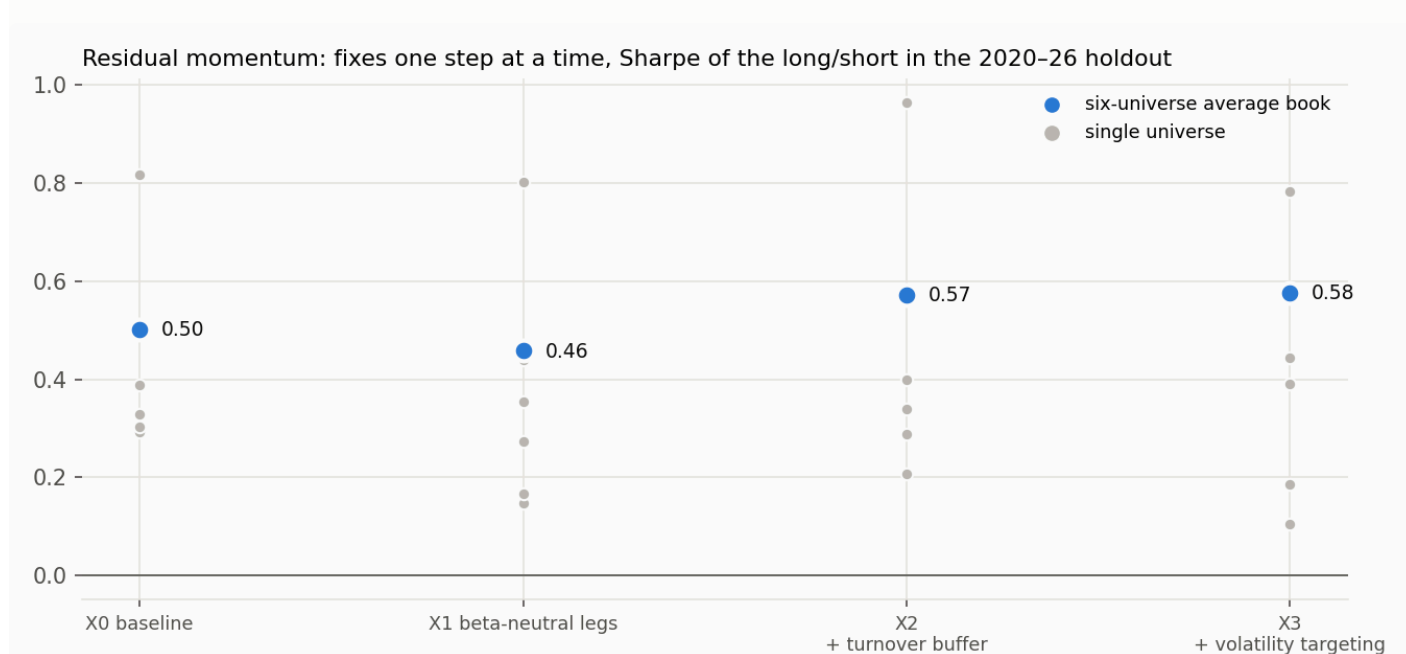
On average across the six universes the gross spread was +3.8% a year, of which the market exposure (beta -0.25) contributed -2.4%; the beta-adjusted spread (CAPM alpha) was +6.3%. Costs took 2.0% a year at 83% monthly turnover across both legs. The long leg beat the universe by +1.8% and the short leg lagged it by +2.0% a year, so most of the spread comes from the short side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages +6.8%, +4.4%, +4.2% a year.

9.2 The fix ladder

X0 baseline → X1 **beta-neutral legs** (each leg scaled by 1/its ex-ante beta) → X2 + **turnover buffer** (enter the extreme 1/q, keep a name until it leaves the extreme 3/q) → X3 + **volatility targeting** (scale the L/S to 10% a year using its trailing 6-month volatility, lagged; Barroso & Santa-Clara 2015; Moreira & Muir 2017).

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X0 baseline	0.08	–	0.50	–	–	+5.8%	-16%	+8.9% (2.4)	2.0%
X1 beta-neutral legs	0.12	+0.09 (p 0.202)	0.46	-0.04 (p 0.618)	2 of 6	+4.5%	-18%	+6.2% (1.6)	2.1%
X2 + turnover buffer	0.14	+0.04 (p 0.387)	0.57	+0.11 (p 0.063)	6 of 6	+4.9%	-13%	+6.4% (2.0)	1.1%
X3 + volatility targeting	0.07	-0.10 (p 0.593)	0.58	+0.00 (p 0.495)	1 of 6	+5.7%	-17%	+7.1% (2.0)	1.0%

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.



- **X1 beta-neutral legs:** +0.09 design, -0.04 holdout; helps in one window and hurts in the other: not reliable.
- **X2 + turnover buffer:** +0.04 design, +0.11 holdout; helps in both windows but does not pass the gate.
- **X3 + volatility targeting:** -0.10 design, +0.00 holdout; helps in one window and hurts in the other: not reliable.

Where it ends: holdout Sharpe 0.50 → 0.58, return +5.8% → +5.7% a year (six-universe book).

10. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.

2. **Currency.** Local currency; World sums local-currency returns without conversion.
3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.
4. **Data gap.** Market-only residuals over 24 months (the paper uses Fama-French 3 factors over 36); stock-level factor loadings outside the US would need data we do not have. The first signals appear in early 2014.
5. **Trials.** Every gated cell is in the factsheet trial ledger; the fix ladder is counted inside FSO3–FS12 (25 trials).

11. Academic references

- Blitz, D., Huij, J. & Martens, M. (2011). Residual momentum. *Journal of Empirical Finance*, 18(3), 506–521.
- Grundy, B. & Martin, J. S. (2001). Understanding the nature of the risks and the source of the rewards to momentum investing. *Review of Financial Studies*, 14(1), 29–78.
- Daniel, K. & Moskowitz, T. (2016). Momentum crashes. *Journal of Financial Economics*, 122(2), 221–247.
- Newey, W. & West, K. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Politis, D. & Romano, J. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- Bailey, D. & López de Prado, M. (2012). The Sharpe ratio efficient frontier. *Journal of Risk*, 15(2), 3–44.
- Jensen, T. I., Kelly, B. & Pedersen, L. H. (2023). Is there a replication crisis in finance? *Journal of Finance*, 78(5), 2465–2518.
- Fama, E. & French, K. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.
- McLean, R. D. & Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1), 5–32.
- Novy-Marx, R. & Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1), 104–147.
- Barroso, P. & Santa-Clara, P. (2015). Momentum has its moments. *Journal of Financial Economics*, 116(1), 111–120.
- Moreira, A. & Muir, T. (2017). Volatility-managed portfolios. *Journal of Finance*, 72(4), 1611–1644.
- Treynor, J. & Mazuy, K. (1966). Can mutual funds outguess the market? *Harvard Business Review*, 44(4), 131–136.

12. Reproduce

code/data.py → signals.py, signals2.py, library.py → xs.py + analyse_xs.py FS11 → build_xs.py FS11.

Metrics use Project1 InvestmentLibrary. Licensed and cached prices are not included.

Appendix

A. Performance in detail

US (S&P 500 members, point-in-time), median 499 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+3.8%	+2.9%	13.8%	0.28 [-0.15, 0.66]	0.35	-20%	48%	1.22	-0.22	+6.5% (2.2)	7.6%
Long-only (net)	+14.7%	+14.3%	16.3%	0.90 [0.49, 1.39]	1.46	-26%	64%	4.50	0.92	+3.7% (2.3)	9.3%
Stocks with the strongest stock-specific trend (gross)	+15.8%	+15.5%	16.3%	0.97 [0.55, 1.46]	1.61	-26%	64%	4.85	0.92	+4.8% (3.0)	9.2%
The weakest (gross)	+8.9%	+7.2%	19.5%	0.46 [0.02, 0.98]	0.56	-32%	60%	2.04	1.13	-4.7% (-2.6)	11.9%
EW universe	+12.0%	+11.3%	15.9%	0.76 [0.32, 1.31]	1.12	-28%	67%	3.64	1.00	–	9.6%
Size-weighted universe	+14.3%	+14.0%	14.5%	0.98 [0.52, 1.55]	1.59	-24%	69%	4.72	0.85	+4.0% (2.5)	8.6%

EU (11 national blue-chip indices, point-in-time), median 262 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+2.9%	+2.2%	12.6%	0.23 [-0.34, 0.81]	0.25	-28%	55%	0.80	-0.28	+5.7% (1.5)	7.6%
Long-only (net)	+12.2%	+11.8%	14.2%	0.86 [0.36, 1.46]	1.42	-22%	60%	3.22	0.86	+3.9% (1.9)	7.6%
Stocks with the strongest stock-specific trend (gross)	+13.3%	+13.0%	14.2%	0.94 [0.43, 1.54]	1.60	-21%	60%	3.51	0.86	+4.9% (2.4)	7.5%
The weakest (gross)	+7.3%	+5.9%	18.3%	0.40 [-0.05, 0.88]	0.52	-37%	56%	1.62	1.14	-3.7% (-1.7)	10.4%
EW universe	+9.7%	+9.0%	14.6%	0.66 [0.21, 1.19]	0.98	-26%	60%	2.72	1.00	–	8.7%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Size-weighted universe	+10.0%	+9.4%	14.0%	0.72 [0.25, 1.24]	1.09	-25%	58%	2.96	0.91	+1.2% (0.8)	8.1%

UK (FTSE 350, point-in-time), median 319 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+2.2%	+1.1%	14.6%	0.15 [-0.33, 0.63]	0.11	-25%	52%	0.63	-0.31	+4.9% (1.4)	8.6%
Long-only (net)	+10.1%	+9.3%	14.8%	0.68 [0.18, 1.29]	1.01	-29%	57%	2.65	0.89	+2.5% (1.2)	9.1%
Stocks with the strongest stock-specific trend (gross)	+11.2%	+10.5%	14.8%	0.76 [0.25, 1.37]	1.15	-27%	58%	2.94	0.89	+3.6% (1.7)	9.0%
The weakest (gross)	+6.0%	+4.2%	19.3%	0.31 [-0.14, 0.81]	0.33	-37%	57%	1.28	1.20	-4.2% (-2.2)	12.1%
EW universe	+8.5%	+7.7%	14.4%	0.59 [0.11, 1.18]	0.80	-30%	58%	2.38	1.00	–	9.2%
Size-weighted universe	+8.2%	+7.6%	13.0%	0.63 [0.12, 1.29]	0.87	-28%	62%	2.43	0.83	+1.2% (0.6)	8.1%

DK (OMXC25, point-in-time), median 20 names, 3 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+6.9%	+5.7%	16.4%	0.42 [-0.13, 1.02]	0.53	-31%	54%	1.41	-0.28	+9.8% (2.2)	9.7%
Long-only (net)	+12.8%	+12.1%	16.0%	0.80 [0.32, 1.28]	1.25	-24%	58%	3.62	0.89	+3.4% (1.6)	9.1%
Stocks with the strongest stock-specific trend (gross)	+13.4%	+12.8%	16.0%	0.84 [0.36, 1.33]	1.33	-24%	58%	3.81	0.89	+4.0% (1.9)	9.1%
The weakest (gross)	+4.4%	+2.3%	20.4%	0.22 [-0.36, 0.82]	0.16	-56%	58%	0.73	1.16	-7.9% (-2.7)	12.6%
EW universe	+10.6%	+9.8%	15.4%	0.69 [0.18, 1.25]	0.99	-33%	65%	2.70	1.00	–	9.2%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Size-weighted universe	+8.9%	+7.8%	16.6%	0.54 [-0.02, 1.16]	0.69	-40%	59%	2.00	0.93	-0.9% (-0.3)	11.0%

SCANDI (OMXC25 + OMXS30 + OMXH25, point-in-time), median 70 names, 5 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+3.4%	+2.6%	13.1%	0.26 [-0.30, 0.79]	0.30	-37%	53%	0.92	-0.20	+5.8% (1.5)	7.2%
Long-only (net)	+14.2%	+13.9%	15.0%	0.95 [0.44, 1.53]	1.64	-21%	60%	3.90	0.91	+3.5% (2.0)	7.8%
Stocks with the strongest stock-specific trend (gross)	+15.1%	+15.0%	15.0%	1.01 [0.50, 1.60]	1.79	-20%	60%	4.14	0.91	+4.5% (2.6)	7.7%
The weakest (gross)	+9.2%	+7.9%	17.8%	0.52 [-0.04, 1.11]	0.66	-46%	62%	1.90	1.11	-3.9% (-1.6)	11.5%
EW universe	+11.8%	+11.3%	14.5%	0.81 [0.28, 1.42]	1.22	-29%	65%	3.16	1.00	–	8.7%
Size-weighted universe	+10.9%	+10.5%	13.9%	0.79 [0.28, 1.35]	1.21	-26%	62%	3.18	0.90	+0.3% (0.2)	8.3%

World (US + UK + EU, point-in-time), median 1078 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+2.2%	+1.6%	11.1%	0.20 [-0.24, 0.66]	0.22	-17%	51%	0.83	-0.24	+4.7% (2.0)	6.4%
Long-only (net)	+11.4%	+10.7%	15.7%	0.73 [0.27, 1.24]	1.07	-25%	64%	3.23	0.90	+2.4% (1.8)	9.8%
Stocks with the strongest stock-specific trend (gross)	+12.5%	+11.9%	15.7%	0.80 [0.34, 1.32]	1.21	-25%	65%	3.55	0.90	+3.5% (2.6)	9.7%
The weakest (gross)	+7.3%	+5.5%	19.6%	0.37 [-0.09, 0.88]	0.43	-40%	58%	1.53	1.14	-4.1% (-3.1)	11.9%
EW universe	+10.0%	+9.0%	16.3%	0.61 [0.15, 1.16]	0.84	-30%	62%	2.58	1.00	–	10.1%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Size-weighted universe	+10.2%	+9.4%	15.0%	0.68 [0.19, 1.25]	0.98	-28%	61%	2.77	0.90	+1.1% (1.2)	9.2%

Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.

Calendar-year returns, L/S (net)

Year	US	EU	UK	DK	SCANDI	World
2014	-2.5%	-18.3%	-13.0%	-14.8%	-4.5%	-6.8%
2015	+21.1%	+11.3%	+5.2%	+14.9%	+12.1%	+8.1%
2016	-4.1%	-2.2%	+7.8%	-1.9%	+11.5%	-1.3%
2017	-1.8%	-7.9%	+9.1%	-6.7%	-10.7%	-1.0%
2018	+4.1%	-11.4%	-2.6%	+23.5%	+4.4%	-2.1%
2019	-1.0%	-4.6%	-12.1%	+38.5%	-4.2%	-3.0%
2020	-5.0%	+2.7%	+2.4%	-23.1%	-27.0%	-1.4%
2021	-3.5%	+6.3%	-4.2%	+9.0%	+12.0%	-3.0%
2022	+21.8%	+13.0%	+17.8%	+33.3%	+18.7%	+22.1%
2023	+3.3%	-1.1%	-2.1%	-2.5%	-7.7%	+6.1%
2024	+1.9%	+15.9%	+1.3%	-10.9%	+17.6%	+5.2%
2025	-10.4%	+19.0%	+13.0%	+10.4%	+11.7%	-4.8%
2026	+19.1%	+11.9%	-3.7%	+21.3%	+9.3%	+5.5%

Calendar-year returns, long-only (net)

Year	US	EU	UK	DK	SCANDI	World
2014	+12.9%	-10.9%	-5.4%	-0.6%	+6.4%	-0.9%
2015	+3.0%	+13.3%	+8.8%	+29.1%	+19.4%	+3.0%
2016	+9.1%	+24.9%	+21.8%	+12.9%	+35.9%	+8.8%
2017	+19.0%	+13.4%	+28.9%	+17.5%	+13.3%	+25.8%
2018	-3.5%	-16.0%	-15.0%	-4.5%	-3.6%	-15.2%
2019	+30.4%	+21.8%	+16.7%	+41.7%	+32.8%	+24.0%
2020	+15.5%	+13.1%	+1.4%	+19.6%	+5.4%	+14.2%
2021	+26.1%	+16.6%	+11.8%	+16.1%	+24.9%	+15.7%
2022	-1.1%	-4.1%	-5.6%	-13.3%	-10.8%	-5.6%
2023	+13.8%	+15.0%	+5.2%	+10.7%	+6.3%	+17.7%
2024	+14.1%	+13.6%	+11.1%	-6.6%	+10.0%	+10.1%
2025	+9.4%	+31.1%	+32.8%	+18.2%	+23.8%	+23.8%

Year	US	EU	UK	DK	SCANDI	World
2026	+35.9%	+26.2%	+13.8%	+22.1%	+18.6%	+21.0%

B. Trading record and current book

The rebalance log per universe is in `results/xs_FS11_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

Universe	Rebalances	Names long / short	Turnover long leg / month	Turnover short leg / month	L/S months positive	Worst L/S month	When
US	150	50 / 50	46%	46%	48%	-10.2%	2020-11
EU	149	27 / 26	44%	45%	55%	-13.1%	2020-11
UK	149	33 / 32	46%	46%	52%	-23.7%	2020-11
DK	149	8 / 7	28%	28%	54%	-18.9%	2020-11
SCANDI	149	15 / 14	38%	37%	53%	-13.7%	2020-11
World	150	109 / 108	46%	46%	51%	-14.2%	2020-11

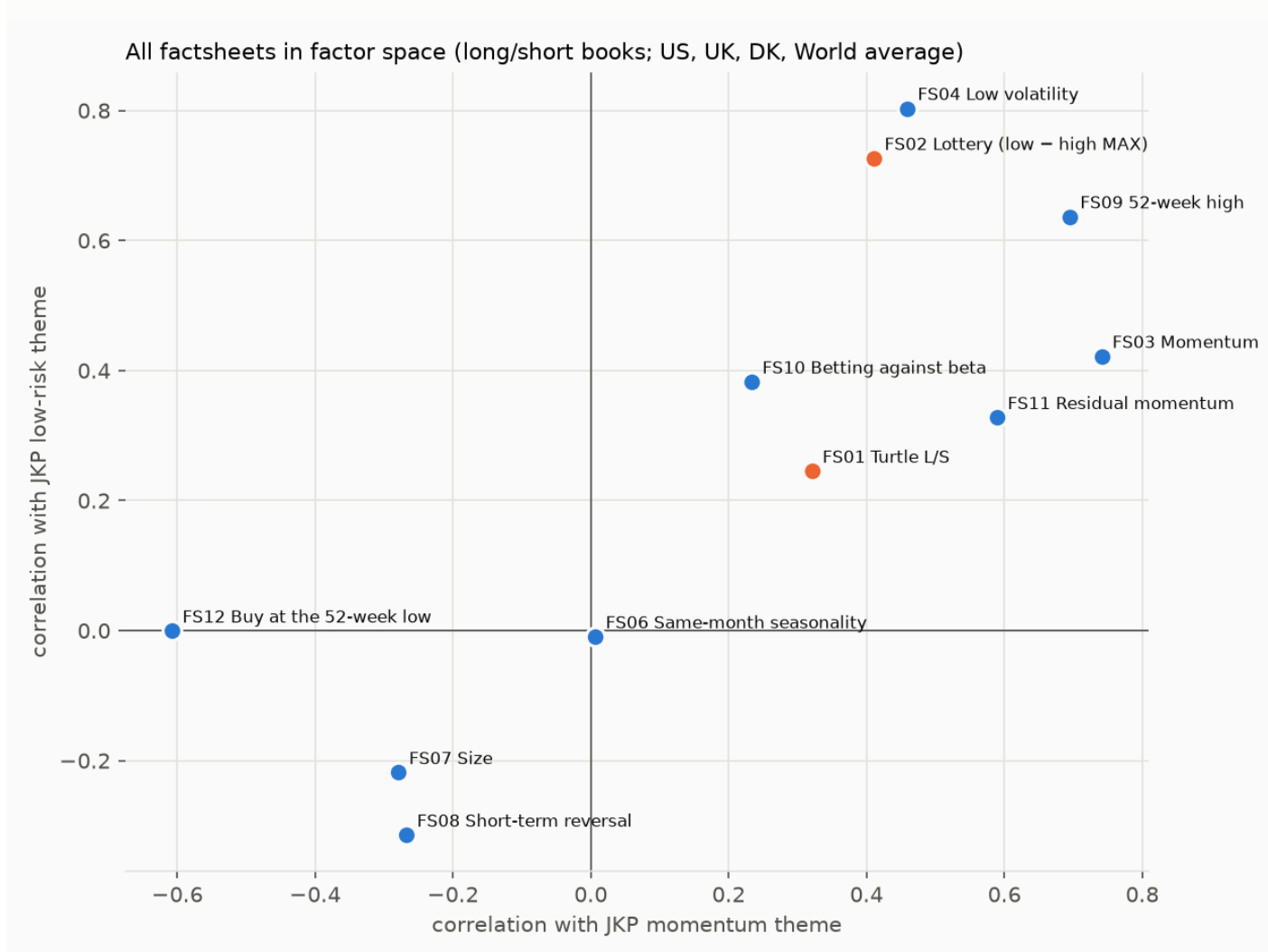
Current book (signal at 31 August 2026; first 8 names per leg)

Universe	Long: stocks with the strongest stock-specific trend	Short: the weakest
US	HST, PFG, FRT, TGT, IEX, EIX, BIIB, CSX	VST, PLTR, UBER, CSGP, NFLX, META, BSX, TSCO
EU	DANSKE.CO, INVE-B.ST, DHL.DE, MRK.DE, SWED-A.ST, INDU-C.ST, NESTE.HE, LOG.MC	BAR.BR, CTT.LS, AGFB.BR, IVG.MI, RMS.PA, G24.DE, CBK.DE, RHM.DE
UK	KLR.L, EMG.L, MRCH.L, BLND.L, RTW.L, CRDA.L, ROSE.L, LAND.L	RNK.L, ENT.L, RMV.L, STJ.L, HWG.L, AVON.L, FCSS.L, AGT.L
DK	DANSKE.CO, NKT.CO, NOVO-B.CO, CARL-B.CO, JYSK.CO, ORSTED.CO, RBREW.CO, VWS.CO	BAVA.CO, TRYG.CO, MAERSK-A.CO, MAERSK-B.CO, COLO-B.CO, NSIS-B.CO, DSV.CO, AMBU-B.CO
SCANDI	DANSKE.CO, INVE-B.ST, SWED-A.ST, NESTE.HE, INDU-C.ST, ATCO-A.ST, TYRES.HE, JYSK.CO	BAVA.CO, MANTA.HE, KEMIRA.HE, SKF-B.ST, KNEBV.HE, VALMT.HE, TEL2-B.ST, ORNBV.HE

C. Classification: where does it fit?

Dimension	Residual momentum
Series family	Momentum & trend
Academic style	Residual (idiosyncratic) momentum
Signal	Price only (daily total returns)
Cross-section vs time series	Cross-sectional: stocks ranked against each other each month

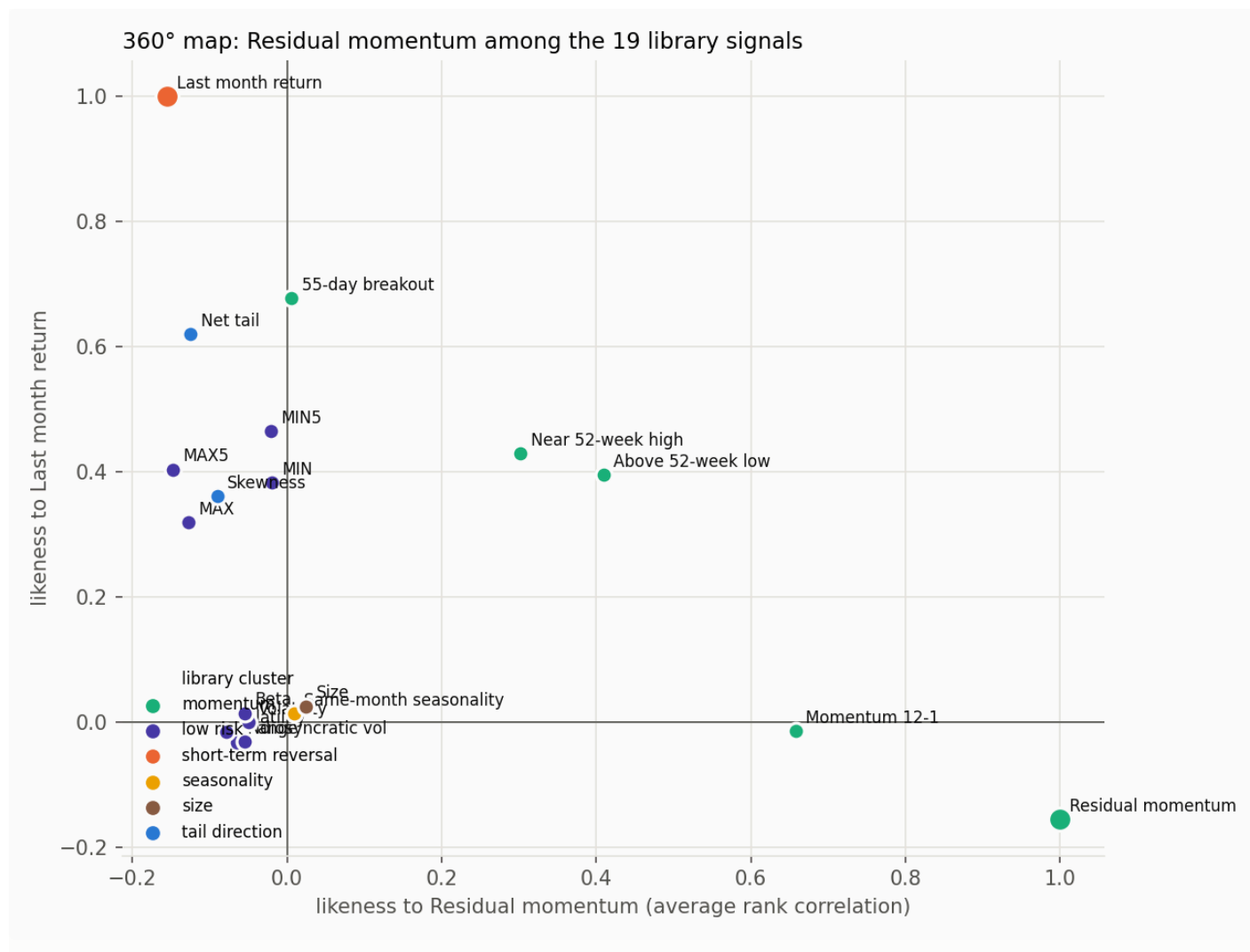
Dimension	Residual momentum
Horizon and turnover	One month; 83% of the two legs replaced monthly
Market exposure	L/S beta -0.31 to -0.20
Payoff shape	Treynor–Mazuy γ -1.66 to -0.25 (t -3.4 to -0.2); worst 10% of market months +0.7% to +3.5% a month, best 10% -2.6% to -0.9%
Nearest factor theme	momentum (4 of 6 universes), WML (2 of 6 universes)
Nearest library signals (returns)	Momentum 12-1 (6 universes), Near 52-week high (6 universes), Above 52-week low (5 universes)
Economic rationale	Same under-reaction story as momentum, but removing the market (and factor) component avoids the time-varying beta that makes plain momentum crash (Grundy & Martin 2001; Daniel & Moskowitz 2016).
Publication	Grundy & Martin 2001; Gutierrez & Pirinsky 2007; Blitz, Huij & Martens 2011
Where it fits	In the momentum cluster of the library; with some information beyond its neighbours.



D. 360° view

The signal among the 19 price signals of the shared library (`code/library.py`), on the same universes and months (equal-weighted top minus bottom quantile, gross). Its natural mirror here is **Last month return**.

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha $t \geq +2$ / ≤ -2
Residual momentum (this strategy)	momentum	+1.00	+1.00	+6.4%	+8.0% to +12.2%	6 / 0
Momentum 12-1	momentum	+0.66	+0.64	+7.2%	+4.3% to +19.4%	5 / 0
Near 52-week high	momentum	+0.30	+0.46	+3.7%	+9.3% to +22.3%	6 / 0
Above 52-week low	momentum	+0.41	+0.44	+7.5%	+5.2% to +13.6%	4 / 0
MAX5	low risk	-0.15	-0.31	+0.6%	-11.3% to -3.5%	0 / 2
Volatility	low risk	-0.08	-0.31	+1.1%	-13.7% to -7.9%	0 / 5
MAX (best day)	low risk	-0.13	-0.29	+0.1%	-13.3% to -2.0%	0 / 2
Range (MAX-MIN)	low risk	-0.06	-0.29	-0.7%	-15.4% to -4.8%	0 / 4
Beta	low risk	-0.05	-0.28	+4.4%	-12.6% to -4.2%	0 / 4
Volatility (252 days)	low risk	-0.05	-0.26	+2.4%	-12.9% to -4.8%	0 / 3
55-day breakout	momentum	+0.01	+0.24	+1.4%	+4.1% to +13.0%	4 / 0
MIN5	low risk	-0.02	+0.24	-0.9%	+2.3% to +13.7%	4 / 0
Idiosyncratic vol	low risk	-0.05	-0.23	-3.6%	-15.5% to -4.8%	0 / 4
MIN (worst day)	low risk	-0.02	+0.23	-0.7%	+3.3% to +11.1%	4 / 0
Skewness	tail direction	-0.09	-0.13	-0.4%	-6.7% to +3.9%	0 / 2
Size	size	+0.02	+0.11	+0.1%	-3.8% to +9.9%	1 / 0
Net tail (MAX+MIN)	tail direction	-0.12	-0.07	-0.9%	-5.2% to +5.2%	0 / 0
Same-month seasonality	seasonality	+0.01	-0.03	-1.9%	-9.4% to +6.6%	0 / 1
Last month return (mirror)	short-term reversal	-0.16	+0.01	+1.8%	-3.2% to +11.0%	2 / 0



Spanning. Regressing the L/S on its five closest library neighbours (by return correlation) and the market:

Universe	Alpha after neighbours / yr	t	R ²	Neighbours with $ t \geq 2$
US	+6.8%	3.54	0.51	Momentum 12-1 +0.39 (+3.3), Near 52-week high +0.31 (+2.2), Size -0.30 (-2.8), 55-day breakout -0.36 (-3.6)
EU	+0.9%	0.29	0.42	Momentum 12-1 +0.42 (+4.4)
UK	+1.8%	0.59	0.51	Momentum 12-1 +0.39 (+3.9), Above 52-week low +0.22 (+2.4)
DK	+6.0%	1.65	0.41	Momentum 12-1 +0.62 (+8.2), MAX5 -0.18 (-2.2)
SCANDI	+4.5%	1.50	0.41	Momentum 12-1 +0.51 (+5.6), Above 52-week low +0.33 (+3.4), Near 52-week high -0.34 (-2.9)
World	+0.8%	0.37	0.61	Momentum 12-1 +0.38 (+4.0), Above 52-week low +0.31 (+3.6), 55-day breakout -0.39 (-4.0), Volatility -0.24 (-4.1)

3×3 sort, next-month return a year
(US, EU, UK, SCANDI, World averaged)

low Residual momentum -	7.8%	7.9%	9.0%
middle -	11.8%	11.0%	9.5%
high Residual momentum -	12.2%	12.4%	12.3%
	low Last month return	middle	high Last month return

Holding Last month return fixed (down a column), moving from low to high Residual momentum changes the return by +4.4, +4.5 and +3.3 points a year; holding Residual momentum fixed (along a row), moving from low to high Last month return changes it by +1.2, -2.3 and +0.0. **Market direction.** Top-minus-bottom return in rising vs falling market months (six-universe average): -1% vs +19% a year. **Horizon:** the gross spread at 1, 2, 3 and 6 months after formation averages +6.4%, +6.8%, +4.4%, +4.2% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)

